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DEMONSTRATION

Reweaving Nishijin: Parametric Design System to Shape the Future of Kyoto Traditional Textiles

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Reweaving Nishijin: Parametric Design System to Shape the Future of Kyoto Traditional Textiles

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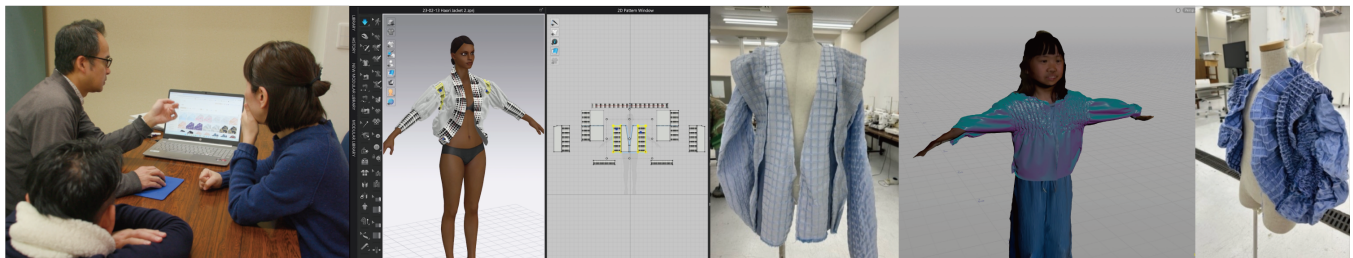


Figure 1: Sharing 3D images through parametric design systems to encourage innovation in traditional communities

Abstract

The Reweaving Nishijin Project revitalizes the industry by combining digital technologies and parametric design. Traditionally used for Japanese attire, Nishijin textiles face declining demand due to competition from Western clothing. This study developed a 3D simulation system to generate garment structures from a single piece of fabric, enabling pre-production design evaluation, reducing waste, and expanding applications. Using Kyoto's Hatcho Nenshi (twisted yarn) and conductive threads, the project produced innovative 3D smart textiles and wearable sensory garments. This fusion of tradition and technology creates sustainable, modern applications for Nishijin weaving. Demonstrations included: 1) A tool for sharing 3D smart textile designs to drive innovation in traditional communities. 2) Displays of high-quality textiles showcasing traditional craftsmanship for visitor interaction.

CCS Concepts

• Human-centered computing → Empirical studies in interaction design.

*Both authors contributed equally to this research.

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Keywords

Weaving, Computer-aided Design, Parametric Design, Smart Textiles, Garment Fabrication, Digital Fabrication

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1 Introduction

Nishijin textile (Nishijin-ori) is a type of textile woven in the Nishijin district of Kyoto. Among Japan's traditional dyed and woven crafts, it is renowned for its exceptional quality and holds the highest reputation. Nishijin textiles are traditionally used in Japan's folk costume, the kimono. However, the kimono industry has declined due to the influence of Western lifestyles, leading to a decrease in sales of Nishijin textile[9].

What sets Nishijin textiles apart is its intricate designs and its luxurious smooth texture, achieved using Japanese-style Jacquard looms. The typical Nishijin textiles are narrow, approximately 40 cm in width, with finely spaced warp threads that allow for intricate weaving. The greater the number of hooks, the finer the woven patterns can be. By combining these techniques with strongly twisted silk threads, such as Hatcho Nenshi and tsumugi threads, used in traditional Japanese crafts, Nishijin textile can produce finely detailed, three-dimensional textiles. These allow for embroidery-like

patterns and enable the creation of the distinctive fine crepe texture characteristic of Tango Chirimen, a traditional silk fabric from Kyoto[30].

Japanese looms feature unique mechanisms, such as the *butou*, similar to the dobby system, and the *fumise*, which lowers the warp threads. The *fumise* mechanism creates large embroidery-like motifs by pressing the warp threads downward, while the *butou* weaves the base fabric, producing textures reminiscent of tweed or satin. Despite their compact size, Japanese looms are capable of crafting textiles on par with those produced by large, computer-controlled Western Jacquard looms, offering precise control over individual threads[30].

Before the advent of digital technology about 40 years ago, Nishijin textiles designs were created manually, with skilled artisans preparing punch cards from design blueprints. However, the development of the CGS format in 1983 by the Nishijin Weaving Industry Association and the Kyoto Textile Research Institute allowed for the digital creation of design data for Jacquard looms[5](Figure2). In this system, the warp threads pass through multiple mechanisms, requiring specialized knowledge during the design process. As a result, the work is often entrusted to skilled, specialized artisans.

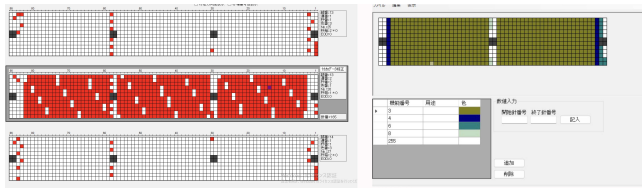


Figure 2: CGS data: the position of the data that each function is linked to is fixed, just as it is when the punch cards were originally used (right), thereby generating the resulting data in a single thread (left).

In Nishijin weaving, much of the design process remains non-digital. The craft is characterized by a highly segmented division of labor, with production progressing through a waterfall-like development process. Currently, the aging of artisans and the decline in their numbers pose significant challenges. To revitalize Nishijin weaving, it is essential to attract and involve new experts in the field. While some businesses are exploring new product development approaches, such as zero-waste design, the limited interaction with experts from other fields hinders the creation of innovative products. It is anticipated that visualizing the design process parametrically in 3D, in a way that is understandable to a diverse range of experts, could increase participation and collaboration.

2 Related Works

In the current Nishijin weaving process, a major drawback is that the design process is understandable only to experts. However, the design process remains opaque and mostly only accessible to specialists. In contrast, Western design software emphasizes the visualization of design workflows, offering user-friendly interfaces with largely standardized operations, albeit at a high cost. Ueda and Milou (2022) were unable to operate a Japanese Jacquard loom using data created with the German software EAT Design Scope due to

differences in design methodologies[30]. To bridge this gap, a new parametric design method is required—one capable of integrating various mechanisms, such as Jacquard systems, knitting needles, and foot pedals, which are traditionally calculated and unified in the minds of skilled artisans to produce final products.

For example, CoilCAM, which utilizes existing parametric design tools such as Grasshopper, has created a new tool designed for use by both artisans and designers[3]. Additionally, AdaCAD, developed by Laura Devendorf et al., features the ability to combine elements within textile software, which also helps improve the design process[4]. Meiklejohn et al. simulated movements such as shrinkage and twisting of active yarns along with weaving structures[18]. There is also a unique study by Asor and Sterman that revisits punch cards to enable parametric knitting[2]. Drawing on these studies, this research tailors the parametric design tool to the features of Japanese Jacquard looms and aims to give back to the traditional craft community of the Nishijin weaving region.

The concept of zero-waste design is critically important in the evolution of fashion. The rectangular shape of most woven fabrics often leads to significant waste during the cutting process. Several studies have addressed this issue by integrating digital design tools[16, 17], and we believe parametric design tools offer potential for minimizing production waste in a similar manner.

Regarding the material value of Nishijin textiles, the Textile Summer School Kyoto, held annually since 2017, has brought together international researchers and designers to explore the possibilities of Nishijin textiles based on Kyoto's traditional dyeing and weaving techniques[6, 24]. Building on the insights from this initiative, our research focuses on utilizing Hatcho Nenshi, a local specialty. The Reweaving Nishijin Project extends these efforts. Generally, three-dimensional textiles can feature multilayered structures, stretchable materials[11, 19, 28, 31], or designs inspired by origami[8, 23]. To create sustainable products from flat textiles, we aim to use Hatcho Nenshi to develop innovative three-dimensional structures while preserving the cultural and material heritage of the region.

Numerous studies have focused on the integration of advanced technologies into the fabrication of smart-textiles[7, 14, 15, 20, 32]. However, woven textiles, particularly those with complex woven structures, have not been sufficiently explored[26, 27, 29]. Most existing studies have focused on incorporating conductive yarns into knitted fabrics[1, 13, 25], leaving a desideratum in the research of woven textiles. This is especially true in Japan, where the development of complex e-textile structures, specifically those woven using Japanese-style Jacquard looms, remains limited. We believe that there is demand in the IT and fashion industries for the development of smart-textile, which is characterized by the fineness of Nishijin textiles.

Research by Kakehi et al. has demonstrated innovative examples of digital product development within Nishijin weaving[10, 21, 22]. However, due to the industry's segmented division of labor, such initiatives have not yet become widespread across the entire Nishijin weaving community. It has been noted that consideration for the community is crucial when digitizing traditional crafts[12]. Therefore, it is essential to explore digitalization approaches that align with the needs and values of the community.

In summary, a review of related studies highlights two key contributions our research makes to the CHI community. First, it provides practical findings on improving the design process within traditional craft production hubs. Specifically, it demonstrates how parametric design tools can make development processes, traditionally restricted by the expertise of artisans, more accessible. Second, it emphasizes the material value of high-quality Nishijin textiles achieved through parametric design tools. This approach expands the range of possibilities for integrating delicate smart textiles into innovative applications.

3 Method

In the Reweaving Nishijin Project, our goal is to integrate technology into traditional crafts in a sustainable manner that respects the artisans' technical heritage. This study aims to establish a design approach that leverages technology to expand the possibilities of Nishijin textiles and facilitate the development of innovative products. To achieve this, we have adopted a parametric design system. This system does not diminish the cultural significance of traditional crafts but instead serves as a tool to make them more accessible to diverse artisans and individuals with fresh ideas.

Parametric design is a methodology that utilizes parameters and elements to create parametric data flows within software, allowing for the visualization of new outputs. For example, it enables iterative experimentation in simulations to create compelling textile designs, which can then be translated into physical products based on the generated data. Because much of the process can be conducted digitally, stakeholders can reach a consensus before production begins, reducing the number of prototypes required and minimizing the labor demands on artisans.

This approach also allows various stakeholders to participate in aspects of the design process that were previously the domain of specialized artisans. It fosters the sharing of creative ideas, potentially leading to more innovative outcomes. While the Nishijin textiles production environment involves many experts, its highly specialized compartmentalization of labor often limits opportunities for mutual exchange. By increasing opportunities for discussion, stakeholders can share their perspectives and collaboratively explore more sustainable and appropriate pathways for the future of Nishijin textiles.

3.1 Parametric Design System for Nishijin

The Parametric Design System for Nishijin weaving focuses on visualizing outputs to facilitate communication. Considering the concept of zero waste, it supports the creation of body-fitting garments by leveraging tools such as Grasshopper, Houdini, Clo3D, and Processing. These tools enable visualization of the outputs while exploring optimal solutions from a diverse range of fabric variations generated with random values.

Physically-based software like Grasshopper and Houdini assists in combining different materials, twisting yarn variations, and manipulating weave structures. While Clo3D supports fashion pattern design, its capabilities for textile design and yarn combination are limited. Conversely, Processing proves useful when paired with Arduino for integrating sensor data. It can utilize the results of

digital measuring tools to adjust patterns for various body types and calculate corresponding fabric designs.

Existing textile software is typically limited to 2D representations, making it less effective for understanding 3D fabric structures or creating complex multilayered patterns. This study utilized a parametric modeling system[8] to develop 3D textiles while considering the unique standards and methods of describing 3D structures specific to Nishijin weaving. The parametric design system incorporates the algebraic logic that weavers use to explain fabric structures to non-weavers[4]. Collaboration with traditional craft unions and the expertise of skilled artisans played a vital role in this process.

3.2 Outline of Implementation

As a validation of the parametric design system in 2020 to 2023, we decided to create a formal jacket (haori), traditionally worn over a kimono for additional warmth. To align with zero-waste design principles, we utilized rectangular textiles, draping them to transform into body-fitting shapes. The textile incorporated a combination of highly twisted threads, silk, polyester fibers, and conductive yarns (resistance of $20 \Omega/cm (\pm 20\%)$).

Using virtual simulations, we predicted the elastic behavior of the highly twisted threads in advance, integrating these findings into the real-world design process. While computational simulations for garments can be implemented using Clo3D, its capabilities for textile design and thread combinations are limited. Therefore, we employed Houdini to replicate the elastic behavior of the highly twisted threads. However, it should be noted that Houdini is currently aimed at visualization purposes and does not yet provide a precise reproduction of real-world physical properties.

After identifying a zero waste pattern, the jacket's template data was imported into Houdini in vector format to simulate garment fitting, similar to Clo3D. For this experiment, we limited our evaluation to square patterns that are well-suited to the loom settings. To simulate fabric stitching and garment fitting on a human model, we utilized Houdini's Vellum system, a particle-based physics simulation framework. In alignment with the actual weaving process, we defined elastic forces (springs) between squares (double-layered fabric) to replicate the behavior of highly twisted threads (elastic yarns) during simulation. This allowed us to reproduce the contraction forces typical of these threads.

The simulation revealed the post-wear design and enabled precise measurements of the completed garment. Consequently, it became possible to predict changes in resistance values when conductive yarns were incorporated, facilitating the simultaneous evaluation of the garment's functionality as a smart textile. Final adjustments to the garment dimensions were made using a digital tape measure integrated with Processing, ensuring accurate and optimized results.

4 Results

4.1 Textiles and Clothes

By employing traditional Nishijin textiles techniques, we produced high-quality textiles with a smooth texture and intricate details. Through experiments using high-twist yarns, silk, and polyester fibers, we developed innovative fabric textures and custom-fit characteristics.

However, we encountered challenges in simulation due to material behaviors that deviated from initial predictions. For instance, the shrinkage rate of polyester high-twist yarns was significantly lower than that of silk high-twist yarns, necessitating pre-testing to incorporate accurate shrinkage rates into the design process. To avoid discrepancies between predicted and actual shrinkage rates, further research is required to fully integrate the specific properties and characteristics of materials into the system.

4.2 Design Process Changes

Previously, the shrinkage behavior of Hatcho Nenshi could only be determined after heat treatment. However, the 3D parametric design system enabled us to predict shrinkage to some extent and facilitated discussions about the anticipated final product with stakeholders before production. Before running simulations in the system, we conducted multiple experiments to align the software's values with the actual shrinkage rates of Hatcho Nenshi, ensuring appropriate parameter settings. While knowing the shrinkage rate of standalone Hatcho Nenshi yarn is helpful, the shrinkage rate of woven textiles varies depending on fabric tension, density, and thickness, preventing perfectly accurate simulations. Currently, textile designers and artisans supplement predictions with their empirical assessments during discussions, but continued efforts to improve simulation accuracy are necessary to better match real-world scales.

Before using the system, it was challenging to predict the behavior of smart textiles. However, the ability to anticipate fabric shrinkage has allowed for predictions of resistance changes. For instance, if the width of a front jacket panel shrinks from 70 mm to 42 mm, the resistance along the same weft direction can be estimated at approximately 1.4 k Ω . This insight is invaluable for planning electronic circuit designs in advance. Additionally, producing small sample pieces for partial testing enabled the creation of highly accurate designs for predicting overall resistance changes. Since the resistance of conductive yarn with high resistivity can be calculated additively based on its length, the system made it possible to achieve precise predictions with increased reliability.

The manufacturing process of Nishijin textiles involves a diverse group of highly specialized stakeholders, each performing their designated tasks. Traditionally, there was little opportunity for these participants to share a unified vision of the final product. However, the introduction of the 3D parametric design system has allowed many artisans to contribute to shared ideas about the potential applications of Nishijin textiles and has provided more opportunities for stakeholders to voice their opinions on its development. While we have not yet achieved dialogue with every participant due to logistical challenges, such as the distance between production sites and the artisans' busy schedules, we believe that the parametric design method has facilitated a greater exchange of honest feedback than previously achieved.

Through this project, the increased opportunities for dialogue enabled the production of textiles with diverse structures, which were uncommon in conventional Nishijin textiles. By leveraging the traditional strengths of Nishijin weaving, stakeholders gained easier access to prototypes featuring intricate designs, which the craft has historically excelled at. This enhanced accessibility has

provided a foundation for innovative product development. Several prototypes were successfully completed, and Nishijin textile union-hosted exhibitions have served as platforms for exchange, fostering spaces within the Nishijin community to discuss new ideas. Moving forward, we aim to continue collaborating with Japan's craft industries to explore new possibilities in manufacturing.

5 Conclusion

Reweaving Nishijin is a project that incorporates digital technology to create new textiles and revitalize the industries in production regions. This study introduces parametric design methods into the Nishijin textiles production process to establish a workflow for the efficient creation of diverse new garments. A 3D simulation system was developed, enabling the generation of three-dimensional garment structures from a single piece of textile on a screen. This system allows all stakeholders to review the design before production, thereby reducing waste and fostering new uses for the textile beyond traditional garments.

By utilizing Hatcho Nenshi, the project successfully developed unique three-dimensional garments. Additionally, textiles incorporating conductive threads were created to enable motion sensing in wearable applications. In the digitalization of traditional crafts, it has been found that sharing concrete 3D representations that are easy for anyone to understand plays a crucial role.

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